RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Department of Civil Engineering

Date: 19-03-2024	**CIE – III (Test)**	**Max. Marks: 50**
Semester: III	**UG**	**Duration: 90 Min**
Course Title: Environment and Sustainability		**Course Code: CV232AT**

S No	Questions	M	BT	CO
1a	Enumerate the impacts of ozone layer depletion.	5	2	3
1b	Explain the concept of 5R.	5	2	3
2	Define the following. i) EIA ii) Climate change iii) Carbon Credit iv) Carbon Footprint v) Green Building.	10	2	3
3	Explain the principles of CSR.	10	2	4
4	Enumerate the ways to reduce carbon foot print.	10	3	3
5a	Briefly explain the concept of circular economy.	5	2	3
5b	Explain the concept of charity and corporate philanthropy.	5	2	4

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Department of Civil Engineering

Date: June 2025	**Improvement CIE (Quiz + Test)**	**Max. Marks: 50 + 10**
Semester: IV	**UG**	**Duration: 90 Min + 20 Min**
Course Title: Environment and Sustainability (Basket Course)		**Course Code: CV242AT**

S No	Questions	M	BT	CO
	Part A			
1.	List the main focus of environmental economics?	1	1	1
2.	What does GDP stand for?	1	2	1
3.	Name one key difference between linear and cyclical resource management systems.	1	3	2
4.	What is the goal of a circular economy?	1	2	2
5.	What does ISO 14000 series focus on?	1	2	1
6.	Describe the significance of carbon credits used for?	1	1	3
7.	What is the meaning of the “R” in zero waste and R concept?	1	1	1
8.	Give an example for cyclical systems.	1	2	4
9.	What is a green building?	1	1	3
10.	Define Environmental Impact Assessment (EIA)?	1	2	4
	Part B			
1.	Explain the concept of sustainability. Discuss the challenges in achieving sustainable development.	10	2	2
2.	Differentiate between linear and cyclical resource management systems. How can cyclical systems contribute to sustainability?	10	2	3
3.	Describe the Sustainable Development Goals (SDGs). Explain any three goals in detail with their targets and indicators.	10	2	4
4.	What is Environmental Impact Assessment (EIA)? Explain its process and importance in sustainable development.	10	2	1
5.	Explain the role of green technology and industrial ecology in achieving sustainable development.	10	2	4

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Department of Civil Engineering

Date: 27/08/2024	**Improvement CIE**	**Max. Marks: 10Q + 50T**
Semester: IV	**UG**	**Duration: 120 Min**
Course Title: Environment and Sustainability		**Course Code: CV242TA**

S No	Improvement Quiz – Questions	M	BT	CO
1.1	Mention any two major causes for climate changes.	2	1	3
1.2	List the benefits of circular economy?	2	1	3
1.3	Discuss energy efficiency, and why is it important in reducing environmental impact?	2	2	3
1.4	List the advantages of Green building.	2	1	4
1.5	Discuss the challenges, when trying to implement the 5Rs?	2	2	4

S No	Improvement Test – Questions	M	BT	CO
1.	Explain the key objectives of conducting an Environmental Impact Assessment (EIA).	10	2	3
2.	Discuss the role of Life Cycle Assessment (LCA) in calculating the carbon footprint of a product.	10	2	3
3.	Explain the principles of Corporate Social Responsibility.	10	2	4
4.	Discuss the history and evolution of Corporate Social Responsibility (CSR).	12	2	4
5.	Discuss the benefits of corporate philanthropy.	08	2	4

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Department of Mechanical Engineering

Date: January 2025	**CIE – Improvement Test**	**Max. Marks: 50 + 10**
Semester: III	**UG**	**Duration: 120 Min**
Course Title: Material Science for Engineers		**Course Code: ME232TB**

S No	Questions	M	BT	CO
	Part A (Quiz)			
1	Mention the main purpose of annealing heat treatment process of ferrous materials.	1	1	3
2	In _____ type of annealing, the lamellar pearlite is transformed into globular type structure.	1	1	3
3	In pack (solid) carburizing _____ is used as energizer.	1	1	3
4	Give two examples for 1D nanomaterial.	2	1	3
5	What happens to the elastic modulus property, when the material brought to nanoscale?	1	1	4
6	Name two top-down methods of nano-material synthesis.	2	2	4
7	In sol-gel process the formation of an oxide network takes place through _____ reactions of a molecular precursor in the liquid.	1	1	4
8	Mesoporous materials are a class of nanomaterials characterized by their highly ordered porous structure with pore sizes ranging from _____ to _____ nanometers.	1	1	4
	Part B (Test)			
1a)	Describe the normalizing heat treatment process with the help of a neat sketch.	05	2	3
1b)	Discuss the flame hardening process with a neat sketch.	05	2	3
2	Illustrate the construction of TTT diagram for eutectoid steel of 0.8% carbon with neat sketches and label various features on the diagram. Also state the applications of the TTT diagram.	10	3	3
3a)	Enumerate the purposes of post-processing heat treatment methods of electronic devices.	05	3	3
3b)	Explain the rapid thermal processing with a neat sketch with its advantages.	05	3	3
4	"Identify whether the chemical vapour deposition (CVD) is top-down or bottom-up approach". Justify your answer and explain the working principle with the help of neat sketch. State its applications and advantages.	10	3	4
5	Discuss the working principle and various modes of atomic force microscope (AFM) with neat sketches.	10	3	4

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Department of Mechanical Engineering

Date: March 2024	**CIE – Improvement**	**Max. Marks: 50**
Semester: 3	**UG**	**Duration: 90 Min**
Course Title: Materials Science for Engineers		**Course Code: ME232AT**

S No	Questions	M	BT	CO
1a.	Define heat treatment and explain the purpose of heat treatment.	5	2	3
1b.	Discuss the advantages and applications of rapid thermal processing for electronic devices.	5	2	3
2a.	Differentiate between annealing and normalising heat treatment processes.	5	3	3
2b.	Describe the full annealing heat treatment process for ferrous materials.	5	3	3
3a.	Explain the steps involved in thermal oxidation of semiconductor devices.	6	3	3
3b.	Describe the defects of heat treatment process for different materials.	4	2	3
4.	With the help of TTT diagram explain hardening process.	10	3	3
5.	Explain the following heat treatment processes with neat sketches: i) Flame hardening and ii) carburising.	10	3	3

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Department of Mechanical Engineering

Date: 03rd June 2025	**CIE – Improvement**	**Max. Marks: 10 + 50**
Semester: IV	**UG**	**Duration: 2 Hrs**
Course Title: Materials Science for Engineers		**Course Code: ME242TB**

S No	Questions	M	BT	CO
	Part A			
1	Mention the purpose of baking in electronic devices.	1	1	3
2	_____ is a process used to grow a thin layer of silicon dioxide (SiO_2) on the surface of silicon wafers by exposing them to an oxidizing atmosphere at elevated temperatures.	1	1	3
3	Highlight any two advantages of Rapid Thermal Processing.	2	2	3
4	Name the heat treatment process variables.	2	1	3
5	Define nanomaterials.	2	1	4
6	Give two examples for top-down approaches of nanomaterials synthesis.	2	1	4
	Part B			
1(i)	Discuss the need of post-processing heat treatment methods for electronic devices.	5	2	3
1(ii)	Briefly explain the steps to be followed in Solder Reflow process.	5	3	3
2(i)	Differentiate Annealing and Normalizing.	4	2	3
2(ii)	With a neat sketch describe induction hardening heat treatment process.	6	3	3
3	Define heat treatment. List and Explain the defects in the heat treatment process.	10	3	3
4	Illustrate with a neat sketch the Chemical vapour deposition technique for the synthesis of nanomaterials.	10	3	4
5	Discuss the working principle of Scanning Electron Microscope characterization technique of nanostructures using a neat labelled diagram.	10	2	4

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Department of Mechanical Engineering

Date: August 2024	**CIE – Improvement**	**Max. Marks: 50 + 10**
Semester: IV	**UG**	**Duration: 120 Min**
Course Title: Material Science for Engineers		**Course Code: ME242AT**

S No	Questions	M	BT	CO
	Part A			
1	Name a critical electronics manufacturing process primarily used in surface-mount technology.	1	1	3
2	Write two key advantages of RTP in Semiconductor manufacturing.	2	1	3
3	What are the two basic purposes of tempering.	2	1	3
4	The crystal structure of martensite is _____.	1	1	3
5	Mention the cause of warpage in heat treatment of mechanical structure.	1	1	3
6	Mention a property of a material which significantly defer from bulk material to nanomaterial.	1	1	4
7	CVD technique used for synthesis of a nanomaterial is a _____ type of approach.	1	1	4
8	How nanoFRP is different from FRP.	1	1	4
	Part B			
1a	Describe the steps followed in thermal oxidation process for the post processing heat treatment of electronic devices.	5	3	3
1b	Highlight the factors affecting thermal oxidation process of electronic devices.	5	2	3
2	Highlight the advantages, disadvantages and microstructures of following heat treatment processes for eutectoid steels: i) Annealing, ii) Normalizing and iii) hardening.	10	2	3
3a	Describe the process of construction of Time Temperature Transformation (TTT) curves of eutectoid steel.	5	3	3
3b	Explain the induction hardening process with a neat sketch.	5	3	3
4a	Define and classify nanomaterials with examples.	5	2	4
4b	Illustrate the synthesis of nanopowder using sol-gel method.	5	3	4
5a	Describe the properties and applications of carbon nanotubes and nano fabrics.	5	3	4
5b	Illustrate the working principle of Scanning Electron Microscope (SEM) for the characterisation of nanocomposites.	5	3	4

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Department of Biotechnology

Date: 8th January 2025	**Improvement Test**	**Max. Marks: 60**
Semester: III	**UG**	**Duration: 120 Min**
Course Title: Bio Safety Standards and Ethics (Basket Course)		**Course Code: BT232AT**

S No	Questions (Quiz)	M	BT	CO
1	What is food processing?	1	1	3
2	What is Innovative Packaging? Give examples.	1	1	3
3	What are the advantages of preservation of food?	1	1	3
4	What are flavouring agents used in food? Give examples.	1	1	4
5	List out the health policies in India.	1	1	4
6	What is food safety?	1	1	4
7	What is ethics in food industry?	2	1	4
8	List out 4 methods used for food analysis.	2	1	4

S No	Questions (Test)	M	BT	CO
1	Classify food preservatives, explain the strategies used for various foods preservation.	10	2	3
2	Illustrate and classify on various food packaging methods used in food industry.	10	3	4
3	"Meat and meat products derived from livestock treated with antibiotics and hormones can cause adverse effects on consumer's health" justify the statement.	10	3	3
4	Elaborate on smart packaging methods with suitable examples. Add a note on packaging materials.	10	4	4
5	Write explanatory notes on: Clinical Ethics and Research Ethics.	5 + 5	2	3

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Department of Biotechnology

Date: 2023-24 (Odd Sem)	**Closed Book Offline Test – 3**	**Max. Marks: 50**
Semester: III	**UG**	**Duration: –**
Course Title: Bio Safety Standards and Ethics (Basket Course)		**Course Code: BT232AT**

S No	Questions (Test)	M	BT	CO
1	Explain on various food processing techniques.	10	2	4
2	Illustrate on the following: Blanching, Pasteurization, Frying.	10	4	4
3	Describe three levels of packaging the food products.	10	4	4
4	Explain the effects of poor packaging on food products, add a note on smart packing.	10	2	4
5	Write Explanatory notes on: Freezing, Dehydration, Homogenization.	10	2	4

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Department of Biotechnology

Date: 3rd June 2025	**Improvement Test**	**Max. Marks: 50 + 10**
Semester: IV	**UG**	**Duration: 120 Min**
Course Title: Bio Safety Standards and Ethics (Basket Course)		**Course Code: BT242AT**

S No	Questions (Quiz)	M	BT	CO
1	List out with two examples smart packing methods.	1	1	3
2	What is GLP? Give the outcomes of GMP in food industry.	1	1	3
3	Why antibiotics are used in food producing animals.	1	1	3
4	What is blanching? Give two advantages of food processing.	1	1	4
5	List out the health policies in India.	1	1	4
6	What are ethics in food industry?	1	1	4
7	What is food production and economics?	2	1	4
8	What are the requirements of Food packing material. Give any two examples of food packing material.	2	2	3

S No	Questions (Test)	M	BT	CO
1	Give an account of various principles and factors of Research ethics.	10	3	4
2	Illustrate on various instruments used in food testing labs for analysis with applications.	10	3	3
3	Elaborate on the principles of clinical ethics with examples; add a note on physician obligations.	10	2	4
4	Elaborate on novel packaging methods with suitable examples.	10	2	3
5	Classify and explain the importance of food additives. Also add a note on the permissible limits for few food products.	10	1	3

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Department of Biotechnology

Date: 27th August 2024	**Improvement Test**	**Max. Marks: 50 + 10**
Semester: IV	**UG**	**Duration: 110 Min**
Course Title: Bio Safety Standards and Ethics (Basket Course)		**Course Code: BT242AT**

S No	Questions (Quiz)	M	BT	CO
1	Explain with two examples about food allergens.	1	1	3
2	What is Research ethics?	1	1	3
3	What are the advantages of preservation of food?	1	1	3
4	What are flavouring agents used in food? Give examples.	1	1	4
5	List out the health policies in India.	2	1	4
6	What is ethics in food industry?	2	1	4
7	List out 4 methods used for food analysis.	2	1	4

S No	Questions (Test)	M	BT	CO
1	Suggest and explain on various strategies involved in food preservation.	10	2	3
2	Illustrate and classify on various food packaging methods used in food industry.	10	3	4
3	"Meat and meat products derived from livestock treated with antibiotics and hormones can cause adverse effects on consumer's health" justify the statement.	10	3	3
4	Elaborate on smart packaging methods with suitable examples. Add a note on packaging materials.	10	4	4
5	Classify and explain the importance of food additives. Also add the permissible limits of few food additives.	10	2	3